

NeDa Eshaghi

Senior AI Engineer/ Data Scientist

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PROFILE

I design and deliver intelligent systems at the intersection of rigorous data science and production-ready machine learning. With 4 years of experience across the full AI life-cycle, from problem framing and data engineering to system design, model development, deployment, monitoring, and continuous improvement. I have built and shipped solutions across diverse domains and high-impact business contexts.

My technical expertise spans Large Language Models, Computer Vision, Speech Processing, and Predictive Modeling, working closely with Product teams to ensure data insights directly shape product strategy and user-facing decisions.

My approach combines structured Error Analysis and rigorous experimentation, with a consistent focus on production impact.

SKILLS

- Machine Learning
- Statistical Modeling
- Computer Vision
- Problem Framing
- Advanced user of Copilot
- Large Language Models
- Recommender System
- Video Processing
- ML System Design
- Big Data Processing
- Data Mining
- Time Series Forecasting
- Speech Processing
- Web Scraping
- Object Oriented Programming

PROFESSIONAL EXPERIENCE

Tapsi, Senior AI Engineer & Data Scientist Chatbot

03/2024 – present | Tehran

- Owned end-to-end design and implementation of a **venture-agnostic chatbot service**, enabling seamless adoption across all Tapsi super-app ventures with minimal integration overhead through a unified, configurable design.
- Achieved 70% inquiry coverage and 80% response accuracy on the **Passenger Support Chatbot** at initial rollout.
- Designed and implemented a Retrieval-Augmented Generation (**RAG**) pipeline leveraging historical customer conversations as dynamic few-shot examples, reducing escalated conversations by **70%**.
- Improved the LLM-based **topic classification model** for **Driver Support Chatbot** conversations, increasing classification accuracy from 20% to 84%.
- Designed and implemented **Camunda-executable conversational workflows** with custom workflow components, enabling configurable chatbot behavior while persisting conversation state, business events, and operational metadata for analytics and auditing.
- Implemented Langfuse-based tracing and observability for LLM applications, enabling end-to-end inspection of prompts, retrieval, tool calls, workflow execution, and model outputs.
- Designed and implemented monitoring dashboards and production metrics to track chatbot quality, response accuracy, escalation rate, workflow health, latency, and operational KPIs.

Robocall

- Improved the Voice Activity Detection model for robocall service through systematic debugging and optimization, increasing precision from 49% to 90%.

Driver Displacement

- Owned end-to-end design and delivery of a real-time driver displacement system to reposition idle drivers from low-demand hexagons to high-yield areas, ensuring displacement viability through proximity and demand thresholds; built an H3-based demand heatmap with high-frequency updates to continuously measure request density and dispatch targeted relocation proposals. Validated end-to-end production implementation, projected 1% uplift in Session Success Rate. Project paused post-launch due to external conditions.

Return Ride

- Designed and deployed a return-ride targeting algorithm by analyzing weekly passenger ride patterns to identify missed return rides, defined spatial eligibility rules for discount allocation, and validated end-to-end production implementation, driving an 8% uplift in rides.

Late Cancellation

- Owned end-to-end design and delivery of a late-cancellation penalty and driver compensation system incorporating cancellation timing, driver displacement, and estimated wait time to minimize platform loss; validated production implementation and ran A/B tests, reducing late cancellations by 3%, achieving 80% penalty recovery rate, and eliminating churn on both sides of the marketplace.

Destination Recommendation

- Led the destination recommendation model enhancement leveraging historical ride data, boosting precision from 57% to 62% and reducing Time-to-Preview by 3%.

Destination First

- Evaluated the Destination First feature, allowing passengers to skip origin selection and book rides faster, through rigorous A/B testing and error analysis, surfacing and resolving critical feature bugs.

Data Warehouse and Monitoring

- Designed and implemented a data warehouse and monitoring dashboard for Tapsi's ride safety team, covering security metrics across all communication channels.
- Implemented end-to-end funnel tracking dashboards for passenger ride requests and driver accepts, providing granular metric visibility across experiments and production rollouts.

Tech Used: PySpark, Python, LangChain, SQL, Metabase, Superset, MongoDB, Poetry, Linux, Git, Docker, Docker Compose, Object Oriented Programming, Qdrant, OpenAI API, LangGraph, Prompt Engineering.

Snapp Market, Data scientist & Machine Learning Engineer

02/2022 – 04/2023 | Tehran

Multi Object Tracking

- Designed and implemented a multi-person tracking system leveraging graph-based modeling across a multi-camera setup, fusing location, appearance, and environmental features to achieve a 35% improvement in tracking accuracy over baseline models
- Developed a model-aided annotation tool for multi-person, multi-camera datasets, significantly improving annotation throughput and labeling quality.
- Developed a real-time camera calibration application for estimating intrinsic and extrinsic parameters.
- Fine-tuned YOLOv7 on the CrowdHuman dataset for detection.

Customer Segmentation

- Designed a customer segmentation algorithm to identify targeted user cohorts, driving a 10% uplift in product bundle recommendation engagement.

ETL Pipeline

- Developed ETL pipelines to automate P&L reporting across teams, enabling stakeholder metric tracking at multiple cadences.
- Automated monthly and quarterly cohort analysis pipelines, unlocking granular performance trend insights across Marketing Team KPIs.

Tech Used: Python, R, PyTorch, OpenCV, SQL, Scikit-learn, Linux, Git, Docker, Object Oriented Programming.

Bonda Innovation Center, Data Scientist / Machine Learning Engineer

02/2021 – 08/2021 | Tehran

VRP

- Architected and implemented a Vehicle Routing Problem (VRP) algorithm to optimize courier delivery routes and costs, achieving a 20% cost reduction in simulation.

Tracking

- Developed and deployed person re-identification and tracking models for production surveillance systems.

Tech Used: Python, Scikit-learn, OpenCV, PyTorch, Linux.

Freelancer, Data Scientist

04/2020 – 01/2024 | Tehran

- Developed a content-based movie recommendation system achieving 4% MSE.
- Conducted statistical analysis to identify optimal locations for digital bank expansion, evaluating transportation costs, demographics, and promotional zones.
- Developed and deployed NSFW detection algorithms on video content for [TILIV.TV](https://tilliv.tv) , optimizing for computational performance.
- Developed a customer segmentation algorithm for an insurance company, improving classification accuracy by 18% over baseline.
- Built a course recommendation system using neighborhood-based collaborative filtering, achieving 87% recommendation accuracy for HR managers.
- Conducted statistical analysis to identify optimal locations for digital bank expansion, evaluating transportation costs, demographics, and promotional zones.
- Developed a scalable web scraping pipeline extracting 1M+ book records to build a book data warehouse.

Tech Used: Python, Git, Docker, Scikit-learn, Flask.

Sadra, Intern Machine Learning Engineer

05/2016 – 09/2016 | Tehran

- Developed and deployed a deep learning model for **autonomous vehicle** speed and steering angle prediction from camera and sensor data, achieving 10% accuracy improvement over baseline.

Tech Used: C/C++.

EDUCATION

Sharif University of Technology

09/2016 – 08/2019 | Tehran

Master of Science (M.S.) in Mechanical Engineering (Control and Robotics)

Final Thesis: Designed active smart shoes to help improve balance for stroke patients.

Sharif University of Technology, Bachelor of Science (B.S.) in Mechanical Engineering

09/2011 – 08/2016 | Tehran

Final Thesis: Designed ML models for multi-agent control, achieving 87% accuracy in speed and position prediction from sensor data.